

App Tutorial

A Simple Nextcloud App Tutorial

Quem sou eu?

Realizador de sonhos desde 2003

Amante de opensource

Palestrante

PHP Zend Certified Engineer ([ZEND024235](#))

PHPRio (<https://telegram.me/phprio>)

CTO LibreCode

Redes sociais: (VitorMattos ou VitorMattosRJ)



Command line tools

OCC

Command line tools

> alias

~/.bashrc

```
if [ -f ~/.bash_aliases ]; then  
    . ~/.bash_aliases  
fi
```

~/.bash_aliases

```
alias occ='docker compose exec -u www-data nextcloud ./occ'
```

Command line tools

> debug

```
occ config:system:get debug  
occ config:system:set debug 1
```

config/config.php

```
'debug' => true,
```

Command line tools

> xdebug

How to use xdebug on VSCode or Codium?

- Open the IDE on folder volumes/nextcloud
- setup the extension PHP Debug or PHP Extension Pack
- Press F5 to start debugging or go to "Run > Start debugging"
- Create a launch.json file to PHP
- Add the follow to your launch.json inside configuration named as Listen for Xdebug:

```
1 "pathMappings": {  
2     "/var/www/html": "${workspaceFolder}"  
3 }
```

Types of apps

apps.nextcloud.com/

App architecture

- **appinfo/**: Contains app metadata and configuration
- **css/**: Contains the CSS
- **img/**: Contains icons and images
- **js/**: Contains the JavaScript files
- **lib/**: Contains the PHP class files of your app
- **src/**: Contains the source code of your vue.js app
- **templates/**: Contains the templates
- **tests/**: Contains the tests

Essentials

> appinfo/info.xml

```
1 <?xml version="1.0"?>
2 <info xmlns:xsi= "http://www.w3.org/2001/XMLSchema-instance"
3
4   xsi:noNamespaceSchemaLocation="https://apps.nextcloud.com/schema/apps/info.xsd">
5       <id>wifi_password</id>
6       <name>Wifi - Password</name>
7       <summary>Create, manage and share wi-fi passwords</summary>
8       <description>Create, manage and share wi-fi passwords using qrcode</description>
9       <version>1.0.0-alfa</version>
10      <licence>agpl</licence>
11      <author>Vinicius Reis</author>
12      <author>Vitor Mattos</author>
13      <namespace>WifiPassword</namespace>
14      <category>files</category>
15      <category>tools</category>
16      <bugs>https://github.com/vinicius73/wifi-password/issues</bugs>
17      <dependencies>
18          <nextcloud min-version="22" max-version="25" />
19      </dependencies>
20 </info>
```

Essentials

> lib/AppInfo/Application.php

```
1 <?php
2
3 declare(strict_types=1);
4
5 namespace OCA\WifiPassword\AppInfo;
6
7 use OCP\AppFramework\App;
8
9 class Application extends App {
10     public const APP_ID = 'wifi_password';
11
12     public function __construct() {
13         parent::__construct(self::APP_ID);
14     }
15 }
```

Essentials

composer init

```
1 {
2     "name": "vinicius73/wifi-password",
3     "type": "project",
4     "license": "AGPL",
5     "authors": [
6         {
7             "name": "Vinicius Reis",
8             "email": "luiz.vinicius73@gmail.com"
9         },
10        {
11            "name": "Vitor Mattos",
12            "email": "vitor@php.rio"
13        }
14    ],
15    "autoload": {
16        "psr-4": {
17            "Root\\Teste\\": "src/"
18        }
19    }
20 }
```

Essentials

composer init

Info: fix the namespace

```
1 {
2     "name": "vinicius73/wifi-password",
3     "type": "project",
4     "license": "AGPL",
5     "authors": [
6         {
7             "name": "Vinicius Reis",
8             "email": "luiz.vinicius73@gmail.com"
9         },
10        {
11            "name": "Vitor Mattos",
12            "email": "vitor@php.rio"
13        }
14    ],
15    "autoload": {
16        "psr-4": {
17            "OCA\\WifiPassword\\": "lib/"
18        }
19    }
20 }
```

Enabling the app

```
occ app:enable wifi_password
```

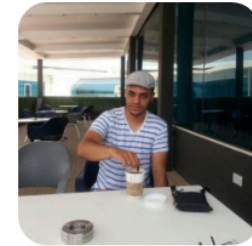
Quality assurance

TDD

Quality assurance

bamarni/**composer-bin-plugin**

No conflicts for your bin dependencies



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Contributors

7k

Used by

530

Stars

27

Forks



<https://github.com/bamarni/composer-bin-plugin>

Quality assurance composer-bin-plugin

```
composer require --dev bamarni/composer-bin-plugin
```

```
1 {  
2     ...  
3     "extra": {  
4         "bamarni-bin": {  
5             "bin-links": true,  
6             "forward-command": true  
7         }  
8     }  
9 }
```

Quality assurance phpcs

```
composer bin coding-standard require --dev \  
nextcloud/coding-standard
```

```
1 "scripts": {  
2     "lint": "find . -name \\.php -not -path './vendor/*' -  
not -path './build/*' -print0 | xargs -0 -n1 php -l",  
3     "cs:check": "php-cs-fixer fix --dry-run --diff",  
4     "cs:fix": "php-cs-fixer fix"  
5 }
```

Quality assurance phpcs

.php-cs-fixer.dist.php

```
1 <?php
2
3 declare(strict_types=1);
4
5 require_once './vendor-bin/coding-standard/vendor/autoload.php';
6
7 use Nextcloud\CodingStandard\Config;
8
9 $config = new Config();
10 $config
11     ->getFinder()
12     ->ignoreVCSIgnored(true)
13     ->notPath('l10n')
14     ->notPath('src')
15     ->notPath('vendor')
16     ->in(__DIR__);
17 return $config;
```

PS: add **.php-cs-fixer.cache** to **.gitignore**

Quality assurance psalm

```
composer bin psalm require --dev vimeo/psalm
composer bin psalm require --dev \
    nextcloud/ocp:dev-master
```

```
1 "scripts": {
2     "lint": "find . -name \\.php -not -path './vendor/*' -
    not -path './build/*' -print0 | xargs -0 -n1 php -l",
3     "cs:check": "php-cs-fixer fix --dry-run --diff",
4     "cs:fix": "php-cs-fixer fix",
5     "psalm": "psalm",
6     "psalm:update-baseline": "psalm --threads=1 --update-
    baseline --set-baseline=tests/psalm-baseline.xml",
7     "psalm:clear": "psalm --clear-cache && psalm --clear-
    global-cache"
8 }
```

Quality assurance

psalm.xml

```
1 <?xml version="1.0"?>
2 <psalm
3     errorLevel="5"
4     resolveFromConfigFile="true"
5     xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
6     xmlns="https://getpsalm.org/schema/config"
7     xsi:schemaLocation="https://getpsalm.org/schema/config
  vendor/vimeo/psalm/config.xsd"
8 >
9     <projectFiles>
10         <directory name="lib" />
11         <ignoreFiles>
12             <directory name="vendor" />
13         </ignoreFiles>
14     </projectFiles>
15     <issueHandlers>
16         <UndefinedClass>
17             <errorLevel type="suppress">
18                 <referencedClass name="OC\*" />
19                 <referencedClass name="OC" />
20             </errorLevel>
21         </UndefinedClass>
22     </issueHandlers>
23 </psalm>
```

Quality assurance

Behat

```
composer bin behat require --dev libresign/nextcloud-behat  
vendor/bin/behat init
```

behat.yml

```
default:  
  autoload:  
    '': '%paths.base%/tests/features/bootstrap'  
  suites:  
    default:  
      paths:  
        - '%paths.base%/tests/features'  
  extensions:  
    PhpBuiltin\Server:  
      verbose: false  
      rootDir: /var/www/html  
      host: localhost
```

Quality assurance

Show me the code



Continuous Integration GitHub Actions

.github



<https://github.com/nextcloud/.github>

Continuous Integration

Show me the code



First page

> appinfo/routes.php

```
1 <?php
2
3 return [
4     'routes' => [
5         ['name' => 'page#index', 'url' => '/', 'verb' => 'GET'],
6     ],
7 ];
```

Routes / API endpoints

connecting controller to route

Name

`authorApi#someMethod`

https://docs.nextcloud.com/server/latest/developer_manual/basics/routing.html

Routes / API endpoints

connecting controller to route

Split at #

and uppercase the first letter of the left part:

```
authorApi  
someMethod
```

https://docs.nextcloud.com/server/latest/developer_manual/basics/routing.html

Routes / API endpoints

connecting controller to route

Append Controller to the first part:

```
AuthorApiController  
someMethod
```

https://docs.nextcloud.com/server/latest/developer_manual/basics/routing.html

First page

> lib/Controller/PageController.php

```
1 <?php
2
3 declare(strict_types=1);
4
5 namespace OCA\WifiPassword\Controller;
6
7 use OCP\AppFramework\Controller;
8 use OCP\AppFramework\Http\TemplateResponse;
9 use OCA\WifiPassword\AppInfo\Application;
10
11 class PageController extends Controller {
12     /**
13      * @NoAdminRequired
14      * @NoCSRFRequired
15      */
16     public function index(): TemplateResponse
17     {
18         return new TemplateResponse(Application::APP_ID, "page-main");
19     }
20 }
```

First page

> templates/page-main.php

```
1 Hello World
```

First page

Add to menu

```
1      ...
2      <dependencies>
3          <nextcloud min-version="26" max-version="26" />
4      </dependencies>
5      <navigations>
6          <navigation>
7              <name>Wifi Password</name>
8              <route>wifi_password.page.index</route>
9          </navigation>
10     </navigations>
11 </info>
```

First page

Add to menu

```
img/app.svg  
img/app-dark.svg
```

Routes / API endpoints

this...

```
1 return [  
2   'routes' => [  
3     ['name' => 'author#index', 'url' => '/authors', 'verb' => 'GET'],  
4     ['name' => 'author#show', 'url' => '/authors/{id}', 'verb' => 'GET'],  
5     ['name' => 'author#create', 'url' => '/authors', 'verb' => 'POST'],  
6     ['name' => 'author#update', 'url' => '/authors/{id}', 'verb' => 'PUT'],  
7     ['name' => 'author#destroy', 'url' => '/authors/{id}', 'verb' => 'DELETE'],  
8     // your other routes here  
9   ],  
10  ];
```

https://docs.nextcloud.com/server/latest/developer_manual/basics/routing.html

API endpoints

can be abbreviated by using the **resources** key:

```
1 return [  
2     'resources' => [  
3         'author' => ['url' => '/authors'],  
4     ],  
5     'routes' => [  
6         // your other routes here  
7     ],  
8 ];
```

https://docs.nextcloud.com/server/latest/developer_manual/basics/routing.html

API Tip



Insomnia

<https://insomnia.rest/download>

<https://github.com/Kong/insomnia>

Getting request parameters

Parameters can be passed in many ways:

- Extracted from the URL using curly braces like **{key}** inside the URL (see [Routing](#))
- Appended to the URL as a GET request (e.g. ?something=true)
- application/x-www-form-urlencoded from a form or jQuery
- application/json from a POST, PATCH or PUT request

Getting request parameters

```
1 <?php
2 namespace OCA\MyApp\Controller;
3
4 use OCP\AppFramework\Controller;
5 use OCP\AppFramework\Http\Response;
6
7 class PageController extends Controller {
8
9     /**
10      * @param int $id
11      */
12     public function doSomething(int $id, string $name='john', string $job='author'):
13         Response {
14         // GET ?id=3&job=Developer
15         // $id = 3
16         // $name = 'john'
17         // $job = 'Developer'
18     }
19 }
```

Getting request parameters

json

```
1 POST /index.php/apps/myapp/authors
2 Content-Type: application/json
3 {
4     "name": "test",
5     "number": 3,
6     "publisher": true,
7     "customFields": {
8         "mail": "test@example.com",
9         "address": "Somewhere"
10    }
11 }
```

Getting request parameters

json

```
1 <?php
2 namespace OCA\MyApp\Controller;
3
4 use OCP\AppFramework\Controller;
5 use OCP\AppFramework\Http\Response;
6
7 class PageController extends Controller {
8
9     public function create(string $name, int $number, string
    $publisher, array $customFields): Response {
10         // $name = 'test'
11         // $number = 3
12         // $publisher = true
13         // $customFields = array("mail" => "test@example.com",
    "address" => "Somewhere")
14     }
15
16 }
```

Getting anything else

```
1 <?php
2 namespace OCA\MyApp\Controller;
3
4 use OCP\AppFramework\Controller;
5 use OCP\AppFramework\Http\Response;
6 use OCP\IRequest;
7
8 class PageController extends Controller {
9
10     public function someMethod(): Response {
11         $type = $this->request->getHeader('Content-Type'); //
12         $_SERVER['HTTP_CONTENT_TYPE']
13         $cookie = $this->request->getCookie('myCookie'); //
14         $_COOKIES['myCookie']
15         $file = $this->request->getUploadedFile('myfile'); // $_FILES['myfile']
16         $env = $this->request->getEnv('SOME_VAR'); // $_ENV['SOME_VAR']
17     }
18 }
```

If you want more, look the interface:
<https://github.com/nextcloud/server/blob/master/lib/public/IRequest.php>

More about controllers

Session
Cookies
and more...

https://docs.nextcloud.com/server/latest/developer_manual/basics/controllers.html

Dependency injection

symfony/**dependency-injection**



Allows you to standardize and centralize the way objects are constructed in your application

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Contributors

 0

Issues

 4k

Stars

 93

Forks



https://docs.nextcloud.com/server/latest/developer_manual/basics/dependency_injection.html

Dependency injection

Don't put new dependencies in your constructor or methods but pass them in.

So this...

```
1 <?php
2 use OCP\IDBConnection;
3
4 // without dependency injection
5 class AuthorMapper {
6     private IDBConnection $db;
7
8     public function __construct() {
9         $this->db = new Db();
10    }
11 }
```

Dependency injection

would turn into this by using Dependency Injection:

```
1 <?php
2
3 use OCP\IDBConnection;
4
5 // with dependency injection
6 class AuthorMapper {
7     private IDBConnection $db;
8
9     public function __construct(IDBConnection $db) {
10         $this->db = $db;
11     }
12 }
```

Middlewares

Middleware is logic that is run before and after each request

https://docs.nextcloud.com/server/latest/developer_manual/basics/middlewares.html

Events

Communicate between different aspects of
the Nextcloud eco system.

Implement a Listener to intercept events

Dispatch events in your app

https://docs.nextcloud.com/server/latest/developer_manual/basics/events.html

Migrations

> occ commands

- **migrations:execute**: Executes a single migration version manually
- **migrations:generate**: Create a new migration file
- **migrations:migrate**: Execute all pending migrations of an app.
- **migrations:status**: View the status of a set of migrations

Migrations

> creating new migration



```
occ migrations:migrate <appid> <version>
```

Migrations

> creating new migration

Migration was named:

Version<version>Date<date>

Example:

Version16000Date20221116163301.php

Migrations

> creating new migration

Version

I recommend to use Semantic Version without dots.

Example:

v1.2.3

will be:

10203

Frontend

> Adding scripts

```
1 <?php
2
3 declare(strict_types=1);
4
5 namespace OCA\WifiPassword\Controller;
6
7 use OCP\AppFramework\Controller;
8 use OCP\AppFramework\Http\TemplateResponse;
9 use OCA\WifiPassword\AppInfo\Application;
10
11 class PageController extends Controller {
12     /**
13      * @NoAdminRequired
14      * @NoCSRFRequired
15      */
16     public function index(): TemplateResponse
17     {
18         Util::addScript(Application::APP_ID, 'page-main');
19         return new TemplateResponse(Application::APP_ID, "page-main");
20     }
21 }
```

Frontend

> Initial state

```
1 class PageController extends Controller {
2     private IInitialState $initialState;
3     public function __construct(
4         IRequest $request,
5         IInitialState $initialState
6     ) {
7         parent::__construct(Application::APP_ID, $request);
8         $this->initialState = $initialState;
9     }
10
11     /**
12      * @NoAdminRequired
13      * @NoCSRFRequired
14      */
15     public function index(): TemplateResponse
16     {
17         Util::addScript(Application::APP_ID, 'page-main');
18         $this->initialState->provideInitialState('config', 'value');
19         return new TemplateResponse(Application::APP_ID, "page-main");
20     }
21 }
```

Frontend

> Initial state

Press F12 and check the input with all initial states

Load in frontend:

https://docs.nextcloud.com/server/latest/developer_manual/basics/frontend/js.html#loading-initial-state

Frontend

> Vue Components



<https://github.com/nextcloud/nextcloud-vue>

Analyzing the code

<https://github.com/nextcloud/app-tutorial>

https://docs.nextcloud.com/server/latest/developer_manual/app_development/tutorial.html

Obrigado!

Redes sociais:
(VitorMattos ou VitorMattosRJ)

